

Creating Positive Social Experiences Through the Design of Custom Wearables

Sabrina Lakhdhir
sabrinalakhdhir@uvic.ca
University of Victoria
Victoria, BC, Canada

ABSTRACT

Wearables are interactive devices that are becoming increasingly popular for tasks such as communication and self-expression. They are used in social contexts by diverse individuals for diverse purposes, and impact social experiences positively and negatively due to their appearance, interactivity, and form. Given that their design impacts social experiences, my thesis considers how wearables can be custom designed by individuals to foster positive social experiences. I take a bottom-up approach to consider how custom designing is driven by individuals' needs, identities and contexts of use, and how custom designing impacts wearable aesthetics and interactions. My thesis contains five projects within the contexts of fashion and healthcare that aim to understand user needs and desires towards custom wearable design, and develop systems and design approaches that empower end-users to engage in such custom design for creating positive social experiences.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Empirical studies in HCI*; *Interactive systems and tools*.

KEYWORDS

customization, wearables, social experiences, bottom-up, DIY, self-expression, making

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1 INTRODUCTION

Wearables can be defined as on-body worn input and output devices that are used in various contexts to help individuals perform routine tasks and communicate information such as health and activity data [16, 37]. They are used for a variety of purposes, ranging from meeting a necessity to expressing oneself [16, 37], and are inherently social - they carry meaning about their wearers, and can be designed to enable or encourage interactions with spectators [36]. As interactive devices that are used in social contexts by diverse

individuals for diverse purposes, the design of wearables can consciously or unconsciously shape social experiences for better or for worse [25, 43, 66]. The concept of social experiences in relation to wearables has previously been associated with end-user needs such as how and what communication takes place [25, 36], and how well devices adhere to social norms [31, 42, 44, 45]. When considering the design of wearables, social experiences are also associated with three considerations that can vary based on user needs and preferences: how they are worn (i.e., their *form*) [33], how they work and are interacted with (i.e., their *function*) [25, 36], and how they are emotionally received (i.e., their *perception*) [44]. For example, in the case of assistive technologies, it is desirable for wearables to be recognizable yet unobtrusive to avoid associated stigmas while ensuring devices are utilized as required [45]. Similarly, we could consider how wearables might be designed differently for interactions with wearers or spectators, or for sharing different types of public or private information [25].

My thesis takes a bottom-up do-it-yourself (DIY) approach to consider how wearable *form*, *function*, and *perception* can be custom designed by diverse users for diverse purposes to create positive social experiences. I consider how individual identities and contexts of use drive custom design, and how resulting aesthetics and interactions are impacted. My methodology is two-fold. My first two projects consist of formative qualitative studies that aim to understand user's desired customizations, how the design of wearables might impact various social factors, how varied motivations and experiences influence desired customizations, and how different individual needs inform how custom design processes are developed. Informed by those insights, my next two projects explore what types of customizations impact social experiences, techniques towards empowering individuals to design better-suited custom wearables, and how we might design technologies that enable diverse end-users to engage in such customization practices. By combining such understandings, my final project aims to highlight relationships amongst user needs, customization practices, and social experiences. By applying varied research methods, I build on literature on wearables and custom design and demonstrate novel approaches towards enabling end-users to ideate unique wearable designs and fabricate custom wearables.

2 RELATED WORK

Wearables are becoming increasingly popular amongst individuals of various demographics for a variety of reasons [16, 37], and they shape social experiences for many individuals across various situations [25, 42, 43]. In human computer interaction (HCI), much of the focus on wearables has been on considering their applications and understanding their design spaces [25, 37, 55]. Social wearables in particular - *on-body worn technologies that augment*

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interactions amongst co-located individuals [25] - have been previously explored through the lenses of social acceptability [42–45, 66], wearability [33, 49, 72], and interaction [25, 42, 49, 65, 66]. Multiple researchers have highlighted the emotional nature of social wearables, experiences, and interactions, demonstrating their complexity and the multidimensional factors that influence their design and use [25, 29, 42–45, 49, 65, 66, 72]. For example, researchers highlight how the social acceptability of devices is dependant on a series of factors including their aesthetics, social image, identity, social norms, ethics, embarrassment, and utility [43, 44]. Similarly, literature demonstrates how wearables can augment individual or social cues, invite interactions, bring people together, and support physical and emotional closeness to facilitate pro-social behaviours [25] through considerations such as its audience and sensing and actuation capabilities [25].

Similarly, customization has been explored within HCI as an approach towards integrating interactions or added functionalities to base devices to make them more personable and relatable for end-users [35, 48, 51, 60, 68]. While such literature does begin to address individual desires to make devices more personalized, such customization efforts still present constraints on what users can and cannot change about their devices.

My thesis considers wearables and custom design as a two-pronged approach towards creating enhanced social experiences. I build upon existing knowledge regarding how wearables look, work, and are received to explore how individual needs, identities, and social norms influence wearable *form*, *function*, and *perception*. I also develop novel techniques to enable diverse demographics of users to engage in custom designing wearable *form*, *function*, and *perception* to understand how such adaptations impact the resulting aesthetics and interactions of wearables and their created social experiences.

3 RESEARCH QUESTIONS

The goal of my dissertation is to study, design, and evaluate novel techniques and systems to facilitate the user-driven design of custom wearables by considering how their *form*, *function*, and *perception* might be altered to create positive social experiences. This overarching goal is composed of three research questions (RQs):

- RQ1:** What types of *form*, *function*, and *perception* changes would individuals like to introduce onto their wearables to create positive social experiences?
- RQ2:** How might we design systems that empower individuals to design custom wearables that account for their form, function, appearance, and interactions in order to create positive social experiences?
- RQ3:** How do the desired and implemented *form*, *function*, and *perception* changes impact individuals' social experiences?

4 RESEARCH APPROACH

To address my identified research questions, I take a human-centered design approach [30] wherein I study, design, and evaluate systems and interaction techniques for supporting the design and creation of custom wearables by iteratively including the end-user in my research, design, and development phases. My research will employ

both, qualitative and quantitative methods to collect data from end-users and assess the tools I develop. Some such methods include workbooks [18, 23, 26, 32], semi-structured interviews [24], and focus groups [58] to understand end-user needs, prototyping [62] for developing suitable systems, and controlled lab studies [51] for evaluating technologies.

I will complete five independent but interrelated projects in the form of case studies [69, 71]. The first pair of projects (1 and 4) consider the case of communicating information through clothing using a series of programmable and non-programmable encodings such as colours, icons, and e-textile components. The second pair of projects (2 and 3) consider the case of customizing glucose monitors for managing type 1 diabetes. These two case studies present a set of similarities and differences, and provide insights towards a breadth of custom design needs and approaches. For example, the first case considers custom design focused on self-expression and communication, where *form* and *perception* are the main driving factors, while the second case considers crucial use cases where *form*, *function*, and *perception* all play integral roles. Both cases introduce unique interaction opportunities and material integrations, and are situated within the domains wherein wearables are said to be most optimistically received by the public [31].

5 RESEARCH TO DATE

5.1 Project One: Understanding Approaches Towards Information Communication Through Clothing

My first project is an empirical study that aims to answer RQ1 and provide insights towards RQ2 from the perspective of hobbyist-sewers who already custom design clothing. Prior research tells us that clothes are an example of wearable artifacts that are worn across various social contexts and have been used as communication devices throughout history [10, 34, 39, 57]. Further, while tools exist to aid the technical aspects of sewing [3, 4, 46, 61, 67, 70], and within HCI to guide the integration of e-textiles [1, 2, 20], there is limited understanding of how individuals can be guided through the creative process of articulating their concepts and ideas for the purposes of self-expression, education, and awareness in social contexts. In this project, I focus on understanding 1) through what *form*, *function*, and *perception*-related alterations individuals aim to communicate information through clothing, and 2) what approaches individuals take towards designing clothes that communicate information while creating positive social experiences.

To gain such understandings, I conducted a three-phase qualitative study with hobbyist-sewers to understand how they design *expressive clothing*, (i.e. clothes that communicate information). I collected data in the forms of sketches [22] (see Figure 1) and interviews [17], and analyzed this data using open and axial coding [5, 50]. From my data analysis, I found that while all participants were interested in embedding meaning into the clothes they created, their approaches towards this integration varied. We identified five unique approaches that our participants undertook when designing *expressive clothing* that address factors such as type of information being communicated and audience of the artifact, as well as a set of four challenges our participants faced in their process. Based on these findings, we discuss a series of implications to inform the

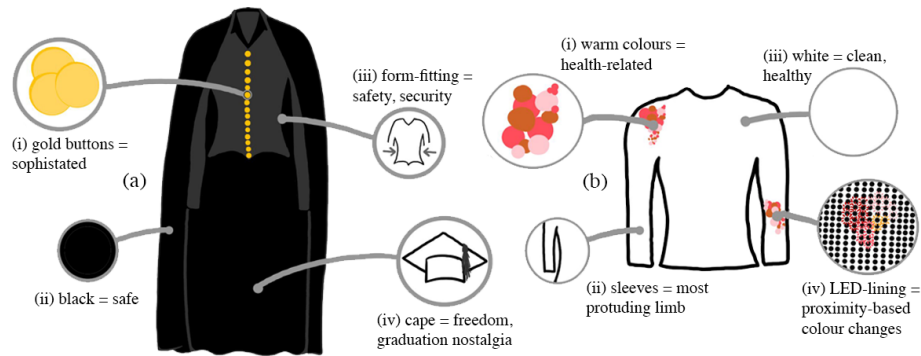


Figure 1: Two examples of participant-proposed *expressive garments*.

design and development of future technologies that can support the creation of *expressive clothing*¹, and further inform my fourth project.²

5.2 Project Two: Towards a Collaborative Tool For Designing Custom Glucose Monitors

My second project is an empirical study that aims to answer RQ1 and provide insights towards RQ2 from the perspective of individuals with type 1 diabetes who use glucose monitors on a daily basis for self-management. Prior research demonstrates that current diabetes monitoring devices present numerous user concerns [7, 8, 27, 28, 53]. Most notably, they are mass-manufactured, bulky, one-size-fits-all devices that do not address user's personal and social needs [9, 41, 56, 63], resulting in the development of DIY communities who find low- and high-fidelity workarounds in isolation [7, 8].

As such isolated DIY workarounds present concerns such as device safety and effectiveness [7, 8], researchers have discussed the benefits of collaborative custom design and proposed end-user inclusive co-design as an approach towards designing better-suited monitors [6, 9, 40, 54, 64]. In this project, I build on efforts by Akyol et al. [7], and aim to understand how end-users imagine their diabetes care technologies functioning, looking, and being perceived by others, and how end-users and professionals envision approaching such a collaborative custom design process.

To gain such understandings, I conducted a three-phase qualitative study that consisted of prototyping, design briefing, and focus group sessions with individuals with type 1 diabetes, healthcare professionals, and product designers.³ Through these three phases, I collected data in the form of images of monitor prototypes [62] made of craft materials and interviews [24], and analyzed this data using deductive coding [19] and thematic analysis [19]. From my data analysis, I identified end-user attitudes towards designing and creating custom glucose monitors, desired customization types, and insights towards how our participants envisioned a collaborative customization design process taking place. Informed by these

findings, we identified a set of five design goals to inform the design and development of future collaborative technologies that can support users in customizing the wearable form and feedback mechanisms of a custom designed glucose monitor. These two forms of customization enable greater consideration of the wearability and social experiences designed devices create. Together, our findings and design goals highlight various parts of our discussed collaborative design process, and provide guidance towards my third project.⁴⁵

5.3 Project Three: *GlucoMaker* Implementation and Evaluation

In this project, I address RQ2 and RQ3 by building upon my findings from project 2 to design and develop an interactive and collaborative system for designing custom glucose monitors. We developed *GlucoMaker*, a web application developed using OpenJSCAD for 2- and 3-D modeling of glucose monitor designs [38]. *GlucoMaker* has five main components: customization guidance interface, device designer interface, collaborative discussion interface, learning portal feature, and project overview feature (see Figure 2a-e, respectively), that enable an end-user to design their own custom glucose monitor. *GlucoMaker* allows end-users to first, select their preferences and adapt or create a custom monitor design, second, collaborate on the digital design by asking questions and engaging in feedback cycles with stakeholders, and third, build monitors using a learning portal page to learn the more technical aspects of design and development.

We evaluated *GlucoMaker* through demonstration [47] by creating three example monitors. In the next phase of this project, I will conduct a user evaluation to understand how *GlucoMaker* supports end-users and stakeholders in designing custom glucose monitors. Through this evaluation, I will gain insights towards how much end-users and stakeholders want to collaborate on the design of custom glucose monitors, and the usefulness of *GlucoMaker*'s main features towards developing custom designs that result in more positive social experiences for device users.

¹This work will be presented at CHI 2024.

²A poster highlighting potential tool development avenues reflective of these results was submitted and presented at a workshop at CHI 2022.

³A discussion regarding challenges of disability inclusive remote co-design was submitted and presented at a workshop at ASSETS 2022.

⁴This work will be presented at CHI 2024.

⁵Before beginning this work, I presented this idea at a federal government-hosted symposium, *Celebrating the Success of Women in STEM*.

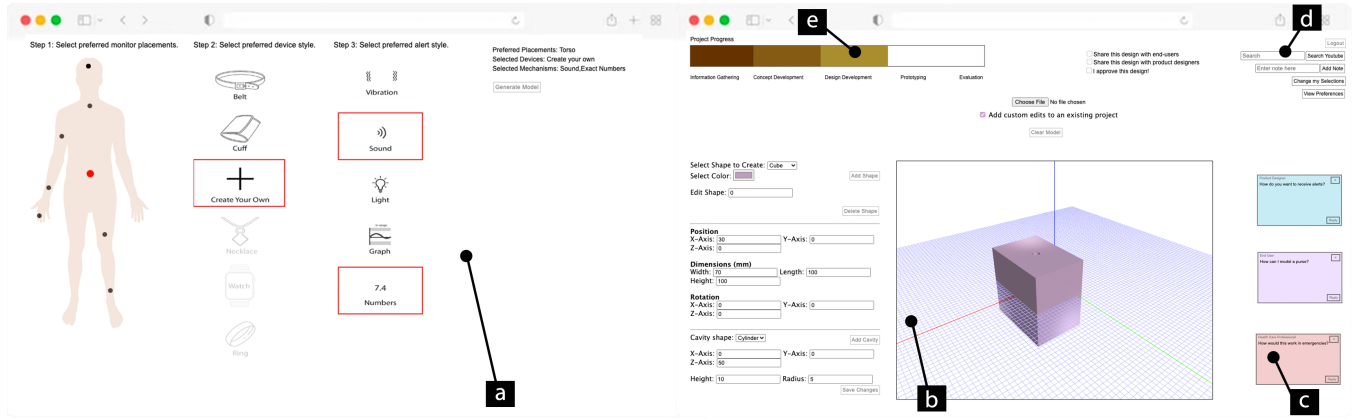


Figure 2: GlucoMaker's five main components: (a) customization guidance interface, (b) the device designer interface, (c) collaborative discussion interface, (d) the learning portal feature, and (e) the project overview feature.

6 PROPOSED RESEARCH

6.1 Project Four: A Technique for Designing Expressive Garments

Informed by my findings from project 1, in this project I will answer RQ2 and RQ3 by developing a hardware- and software-integrated technique that enables users to employ custom clothing design approaches. To employ the five previously-identified design approaches, we have identified a set of properties that must be supported. These include: 1) designing unique data representations, 2) allowing for growth or changes overtime, 3) enabling multiple interaction states, 4) altering the visibility of data, and 5) creating visually appealing and wearable designs. Informed by digital fabrication literature [11–15, 21, 52], I will design and evaluate a new fabrication technique that will enable sewists to fabricate interactive encodings onto their textiles for communication purposes. Following development and validation of this technique by demonstration [47], I will host a workshop with a group of hobbyist-sewers wherein they will be introduced to the technique and provided with an opportunity to design and develop an *expressive* garment utilizing it. Through the proposed work for this project, I will gain insights towards what challenges are faced when software tools are introduced into a primarily tactile workflow, how novel interaction and making techniques are welcomed by and can be taught to individuals who may have limited fabrication knowledge, and how the inclusion of digital fabrication and interactive encodings can enable communication and interactions while adhering to social norms.

6.2 Project Five: Towards an Understanding of Engaging End-users in Custom Designing Wearables

In my final project, I will combine findings from my previous projects to provide a holistic overview encompassing RQ1-3. I will develop an overarching design space regarding end-user driven wearable authorship as it relates to creating more positive social

experiences. Such an overarching project would be situated within both my case studies and my newly identified considerations for wearables within the contexts of device *form*, *function*, and *perception*.

7 FUTURE WORK AND CHALLENGES

Building upon my dissertation research, in the future I will focus on the development of online communities through which individuals can engage with custom design and creation on a deeper level. Such future work would enable the sharing of custom designed wearables, and provide peer support to individuals who may be less familiar or comfortable engaging with technology and self-authored design. I envision such communities recognizing wearables' utilities, and introducing a shift towards collaborative designing for social contexts wherein devices enrich social experiences.

One challenge I anticipate facing is designing technologies to enable users with differing engagement and knowledge levels to partake in custom design and creation processes. Prior literature highlights the various levels of creativity [59] individuals encompass, and I envision leveraging such understandings to design novel features that result in technologies that are more accessible and approachable for broader demographics of users. For example, systems might encourage collaboration amongst expert and novice users, or aim to engage individuals from different domains to share expertise.

8 CONTRIBUTIONS

My research aims to understand how end-users can be empowered to design more positive social experiences by custom designing wearables that evoke greater personal meaning and enable self-expression. My dissertation will contribute:

- (1) An understanding of user needs that describe motivations and approaches towards custom designing the form, function, appearance, and interactions of wearables,
- (2) Recommendations for how tool(kit)s can be designed to support individuals in designing custom wearables that consider

wearable *form*, *function*, and *perception* in relation to user's personal and social needs,

- (3) A set of tools that demonstrate unique approaches towards supporting individuals in designing custom wearables that result in the creation of positive social experiences, and
- (4) A design space that can inform future researchers, designers, and developers on how technology can be designed to encourage diverse individuals with varied needs to create custom wearables that enhance their social experiences.

Together, these contributions will emphasize the importance of designing custom wearables to address individual needs and enhance social experiences, enabling individuals to engage with information, software, and hardware for the purposes of creativity, pleasure, utility, or expressiveness. These contributions to the HCI community will nudge future research towards further supporting custom design and creation processes by demonstrating the potential of end-user driven approaches that enable diverse users to address their needs, and create more positive social experiences. Such efforts strive to make it easier for individuals to engage in such activities, therefore democratizing such practices.

9 RESEARCH STATUS AND LONG TERM GOAL

I am a third year computer science PhD candidate at the University of Victoria, working with Dr. Sowmya Somanath. I successfully completed my candidacy exam (research proposal presentation) in November 2023 and aim to defend my thesis in December 2025. Canadian PhD programs usually span 4-5 years. I have not previously participated in a doctoral consortium.

My long-term goal includes continuing as an HCI researcher and exploring opportunities in industry and academia. I want to continue exploring how individuals can be encouraged, empowered, and enabled to take authorship and ownership over their belongings by designing and creating custom, better-suited wearables. I also hope to continue understanding and evaluating approaches towards the democratized use of technology for more diverse and more accessible engagement with tangible artifacts that impact and improve individuals' day-to-day lives and experiences.

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